

1

Pay attention during the grad student guest lectures during the last week of class. Select one, and summarize the research being conducted by that student. Then, describe the epidemiologic concepts you learned in this class that are being utilized in that student's research. (1-2 short paragraphs will suffice)

Lent's research was titled "Predictors of two-months culture conversion among adult index TB cases in Kampala, Uganda". He spent the first bit of the presentation discussing what is TB and determining the severity of TB. They analyzed how Bandim TB score and Chest X-ray Features predict the two-months culture conversion. Lent looked at a variety of characteristics and used a t-test to determine if the characteristics had different means for if culture converged vs did not culture converged. Lent used LASSO regression to create optimal models for the Chest X-RAY and Bandim TB score. However, the models were not very good as the AUC was roughly 0.55 for both models meaning the models were slightly more accurate than a coin flip but not very. The models also had poor AIC and BIC indicating a very poor fit. Lent found that the most dominant predictor was the BaselineSmearGrade while disease extent was the second highest dominant predictor despite not being statistically significant. Using the models, they concluded that the Bandim TB score and the Chest X-ray are weak predictors of TB.

One of the things covered was the sufficient cause and necessary causes in diseases. Lent said that exposure to *M.tuberculosis* is a necessary cause but is not sufficient cause for development for TB. Lent also used a variety of statistics we learned such as Odds Ratios, Risks Ratios, Risks etc which he used two sample t-tests to determine if the difference was significant.

2

Read the section on chronic disease epidemiology in Chapter 12 (pages 241-242) (Make sure you are using the 3rd edition of the text – this section is not present in the old versions!). Define chronic disease epidemiology and think of one application or area of study of particular public health significance. Why is that area of particular public health concern? (After reading the text, do not feel restricted to the examples given there, you can and should think about this more broadly. Support your point about public health significance with a statistic from an internet search. Again, 1-2 paragraphs should suffice.) (10 points)

Chronic disease epidemiology is the study of the impact of non communicable and chronic diseases on overall population health. Non communicable diseases are often characterized by their long duration and unlike infectious diseases, which are typically caused by a virus or bacteria, chronic diseases are caused by complex web

of causation. Chronic disease epidemiology often explores this web attempting to determine what genetic, behavioural, social and environmental factors play a role in a person's risk of getting the chronic disease. One particular area of study of chronic disease epidemiology is the study of Type 2 Diabetes. Type 2 Diabetes is an important area of study for chronic disease epidemiology because it often acts as a gateway to other serious health issues. In fact, the CDC website on diabetes¹ states that in 2021 diabetes was the 8th leading cause of death in the US with over 100,000 death certificates listing diabetes as the underlying cause of death. Additionally, a crude estimate for the number of people in the US who have diabetes was 38.4 million² in 2021, thats roughly 11.6% of the US population.

¹<https://www.cdc.gov/diabetes/php/data-research/index.html>

²<https://www.cdc.gov/diabetes/php/data-research/index.html>